

Documentation: CSF Neurodegeneration Markers in DZNE studies

Version 1.0. 2023.03.31.

- **Background:**

- Until 2022, CSF amyloid (A β) and tau isoforms were determined for DZNE studies (e.g. DELCODE, DESCRIBE) using manual ELISA methods established at Michael Heneka lab, DZNE Bonn.
- Consistency of methods, using same assay kits and same trained personal at a central lab, was considered crucial to obtain homogenous and comparable data throughout course of studies.
- In 2022, different developments coincided, resulting in a need to change the method used:
 - Advances in automated assays had reached the point where all desired markers could be determined on fully automated platforms standardized to the high quality of clinical diagnostics. It was evident that in the future, this degree of quality would become gold standard for continuous biomarker measurements used through the course of studies.
 - Structural changes of the Heneka lab resulted in loss of trained personal
 - For one of the manual assays (MSD t-Tau), despite advertised degree of standardization, the manufacturer changed the antibody pair used in the assay, as the original antibodies were not available anymore. For this assay, a bridging test for a follow-up assay was therefore unavoidable.
 - Funding of the German national dementia strategy program became available in the framework of the “PreBeDem” grant, enabling acquisition of new instrumentation.
- A Fujirebio Lumipulse G600II system was purchased for the DZNE Bonn Clinical Research Platform (CRP) lab as this system provided highest standardization for all assays among competitors.
- A bridging experiment was designed to re-analyze a limited number of samples – leftover aliquots of previous DELCODE CSF studies – on the Lumipulse system to enable comparison to the manually acquired values.
- The bridging was embedded in the framework of the PreBeDem grant subprojects, which included determination of the CSF neurodegeneration markers in approx. 300 subjects of running DZNE studies.

- **CSF Markers included in the neurodegeneration (“Demenzmarker”) panel:**

- A β 40, A β 42, t-Tau, p-Tau-181, Ratio A β 42/40, Ratio A β 42/p-Tau-181
- The A β 38 protein included in the manual method was not bridged (not available on Lumipulse, limited interest by researchers)

- **Performance documentation of the manual method (retrospective evaluation):**

- Aliquots of a pooled CSF sample (internal control, IC) were used to monitor run / batch variance
- 1 Aliquot run on each of the manual ELISA 96 well plates
- Data collected for mean value and variance analysis from DZNE study runs of 2018 – 2022
- Data from different LOTs, 1 – 11 plates / LOT
- Total of 45 Data Points, each representing 1 run / assay plate
- Total of 6 Analysis batches (5 for t-Tau as MSD could not purchase antibody anymore)

Marker	Assay	Total Batches	Total LOTs	IC Global Average Result	Global CV (%)	Average within Batch CV (%)	Average Duplicate CV (%)
A β 40	MSD VPLEX	6	6	5843 pg/ml	5.4	5.0	2.6
A β 42		6	6	365 pg/ml	10.5	7.4	2.5
t-Tau		5	6	495 pg/ml	26.1	NA	4.6
p-Tau-181	Fujirebio Innotest	6	6	66.7 pg/ml	6.7	NA	1.3
A β 42/40	Ratio	6	6	0.062	9.4	5.7	NA
A β 42/p-Tau-181	Ratio	6	6	5.50	14.1	8.1	NA

- t-Tau assay was the only assay where inter-batch variance exceeded 20%
- All other assays were in acceptable range (though not without variance)
- Within Batch variance (inter-run variance within a LOT/Batch) in acceptable range, between global and duplicate CV
- Technical duplicate variance was always lower than inter-batch variance

- **Overview of DELCODE samples used for re-analysis on the Lumipulse automated system:**
 - Samples that had been originally analyzed by manual method where reanalyzed on Lumipulse system using leftover sample from other DELCODE research projects
 - As leftover volumes were used, not all markers could be obtained at once from single samples
 - Sample 1-10: Combination of A β 40 and A β 42, A β 42/40 ratio, primary rationale to obtain 10 ratios
 - Sample 11 – 20: tTau and pTau 181, primary rationale to obtain 10 values
 - Sample 21 – 30: A β 42 and pTau181, primary rationale to obtain 10 A β 42/pTau181 measures
 - Sample 31 – 40: A β 40 and t-Tau, primary rationale to increase t-Tau values >10 to improve correlation modelling
 - Resulting N = 20 for each single analyte
 - Resulting N = 10 for each ratio

Measurement ID	Aβ40 Manual	Aβ40 Lumipulse	Aβ42 Manual	Aβ42 Lumipulse	t-Tau Manual	t-Tau Lumipulse	p-Tau Manual	p-Tau Lumipulse	Aβ42/40 Manual	Aβ42/40 Lumipulse	Aβ42/p-Tau Manual	Aβ42/p-Tau Lumipulse
1	5315	5285	518	437	Not used				0,097	0,083	Not used	
2	8953	11562	257	315	Not used				0,029	0,027	Not used	
3	10909	11067	478	380	Not used				0,044	0,034	Not used	
4	2951	3033	122	105	Not used				0,041	0,035	Not used	
5	7936	8610	867	642	Not used				0,109	0,075	Not used	
6	18184	15489	1826	1335	Not used				0,100	0,086	Not used	
7	4146	2963	365	240	Not used				0,088	0,081	Not used	
8	16767	13736	1692	1316	Not used				0,101	0,096	Not used	
9	9070	10784	400	386	Not used				0,044	0,036	Not used	
10	4135	5140	141	155	Not used				0,034	0,030	Not used	
11	Not used				683	853	88,6	83,4	Not used			
12	Not used				207	159	24,6	20,6	Not used			
13	Not used				353	358	60,1	51,8	Not used			
14	Not used				628	510	86,5	68,2	Not used			
15	Not used				263	249	42,1	47,8	Not used			
16	Not used				390	367	56,9	60,1	Not used			
17	Not used				552	434	70,2	76,4	Not used			
18	Not used				994	674	128,0	108,7	Not used			
19	Not used				426	562	72,3	64,9	Not used			
20	Not used				329	249	46,6	36,5	Not used			
21	Not used		560	504	Not used		27,7	23,7	Not used		20,23	21,27
22	Not used		393	351	Not used		40,2	31,5	Not used		9,79	11,14
23	Not used		1012	867	Not used		36,9	41,6	Not used		27,43	20,84
24	Not used		802	636	Not used		42,7	30,1	Not used		18,77	21,13
25	Not used		665	418	Not used		59,0	53,6	Not used		11,27	7,80
26	Not used		832	630	Not used		59,6	47,9	Not used		13,97	13,15
27	Not used		282	249	Not used		70,9	57,4	Not used		3,97	4,34
28	Not used		425	317	Not used		68,0	48,3	Not used		6,25	6,56
29	Not used		428	411	Not used		101,2	123,5	Not used		4,23	3,33
30	Not used		244	221	Not used		126,8	162,9	Not used		1,93	1,36
31	5210	5726	Not used		226	176	Not used					
32	4700	5332	Not used		352	310	Not used					
33	12044	11885	Not used		399	270	Not used					
34	9837	12326	Not used		466	279	Not used					
35	9966	11895	Not used		520	273	Not used					
36	6766	7557	Not used		567	342	Not used					
37	9837	12994	Not used		708	590	Not used					
38	7131	7410	Not used		908	571	Not used					
39	11334	13844	Not used		1157	1106	Not used					
40	10125	11970	Not used		1406	1093	Not used					

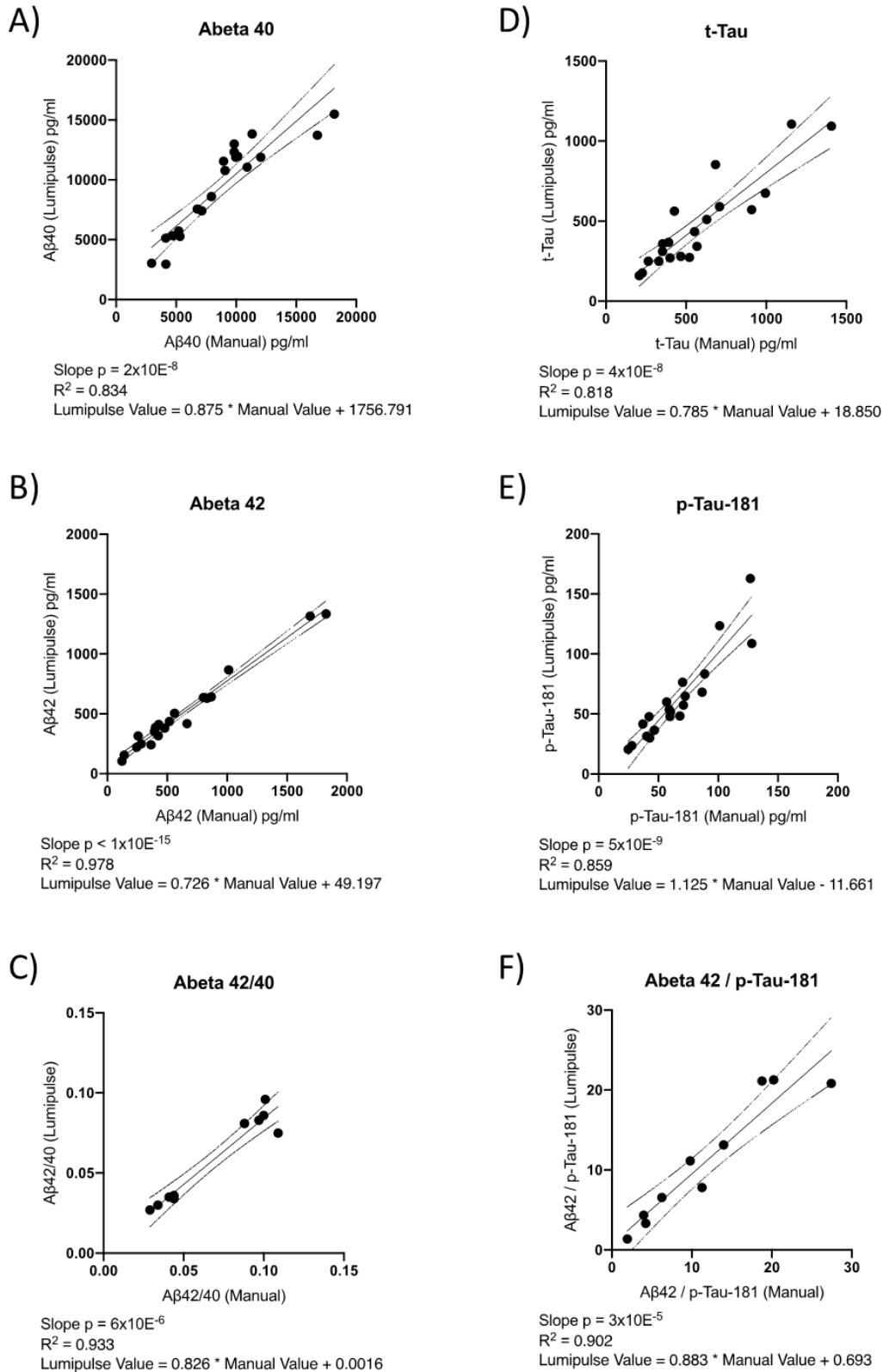


Figure 1: Correlation between manually and Lumipulse measured values.

Spearman correlation with linear fit and 95% confidence interval based on the 10-20 bridging values obtained by reanalysis of samples on Lumipulse system in 2022. Linear fit shows high correlation coefficients and presents a useful model to bridge between systems. Therefore, a simple linear function can be used to transform manually derived values to match Lumipulse values (or vice versa).

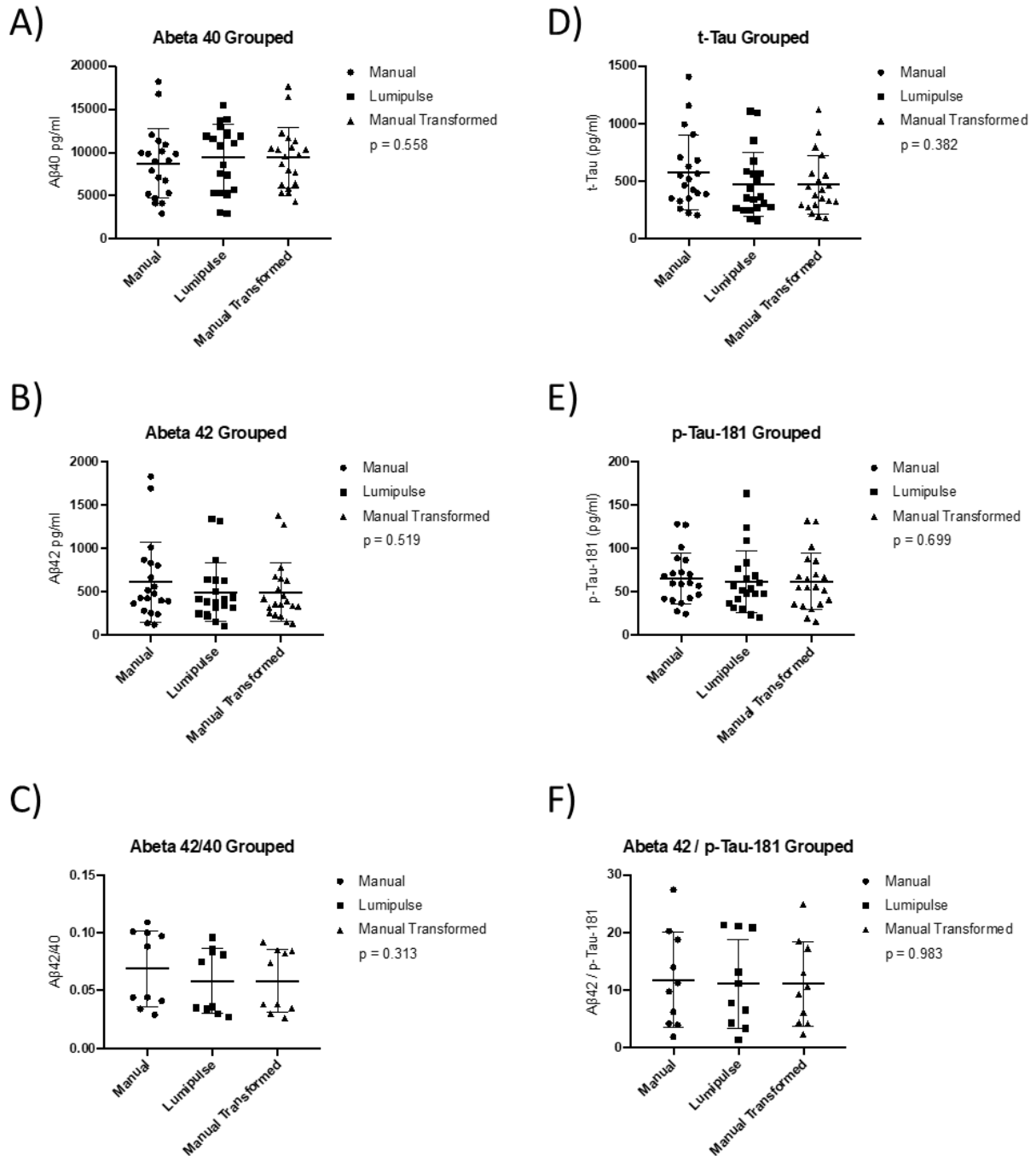


Figure 2: Comparison of manual, transformed and Lumipulse derived values.

The figure displays values obtained by manual method next to values of the same sample set obtained by Lumipulse system. At the right site, values derived by manual system but transformed using the functions displayed in figure 1 are shown. Depending on the analyte, there is a trend to small differences between values derived from the different systems. Transformed manual values are matching Lumipulse values, indicating usability of the transformation for joint analysis of data obtained from both the old manual and the new Lumipulse system.

- **Available data and organization in the database**

- All CSF dementia markers (amyloid and tau isoforms) have been obtained with either the manual or the automated method.
- No samples have a mix of data from the independent methods in the database.
- An assay variable “ND_Assay” indicates for each sample the method that has been used: 0 = Manual method, 1 = Lumipulse. This is therefore valid for all of the analytes.
- For both methods, amyloid isoforms A β 40, A β 42, A β 42/40, t-tau, p-tau-181, and A β 42/p-Tau-181 ratio are available.
- Only for the manual method, A β 38 data is available.
- Only for the manual method, coefficient of variance (CV) in % is reported. The quality control criterion was that the CV was <20%.
- For the Lumipulse system, due to the performance of this automated system, there is no need to run technical duplicates and therefore no CV is reported.
- Run quality controls, like technical inter run controls, are monitored by the team during acquisition of data but not documented in the database. Only data of runs without technical failures are reported in the database.

- **Transformation functions:**

- If data of both analysis systems is used in one analysis (instead of splitting it by assay variable), use the following transformation functions to match data before analysis:
 - **A β 40)** LUMIPULSE VALUE = $0.875 \times \text{MANUAL VALUE} + 1756.791$
 - **A β 42)** LUMIPULSE VALUE = $0.726 \times \text{MANUAL VALUE} + 49.197$
 - **A β 42/40)** LUMIPULSE VALUE = $0.826 \times \text{MANUAL VALUE} + 0.0016$
 - **T-Tau)** LUMIPULSE VALUE = $0.785 \times \text{MANUAL VALUE} + 18.850$
 - **P-Tau-181)** LUMIPULSE VALUE = $1.125 \times \text{MANUAL VALUE} - 11.661$
 - **A β 42/p-Tau-181)** LUMIPULSE VALUE = $0.883 \times \text{MANUAL VALUE} + 0.693$
 - ***Please note that follow-up tests to improve the bridging modelling are under discussion, and that the details of the modelling might receive updates as more data is generated on the Lumipulse system.***

- **Cut-off Values and Citation**

- For the data obtained by the manual method, description of the method and principle of cut-off value determination are described in the following publications:
 - Jessen et al., Design and first baseline data of the DZNE multicenter observational study on predementia Alzheimer’s disease (DELCODE). <https://alzres.biomedcentral.com/articles/10.1186/s13195-017-0314-2>
Cut-off values for the manual method were based on Gaussian mixture modeling of the DELCODE data independent of any group assignments using the R package flexmix (version 2.3-15) (Jessen et al., 2022): <https://alz-journals.onlinelibrary.wiley.com/doi/10.1002/alz.12674>
- Cut-off values of manual and Lumipulse method:
The table below lists for each analyte the cut-off value for pathological range, based on the data-driven cut-off determination used for the manual method. For the Lumipulse system, there was no DZNE study specific evaluation of cut-offs

values yet as data needs to be generated first. Instead, cut-offs for the manual method were transformed using the functions displayed in figure 1.

Method	A β 42	A β 42/40	t-Tau	p-Tau-181	A β 42/p-Tau-181
Manual	≤ 638.7	≤ 0.08	≥ 510.9	≥ 73.65	≤ 9.68
Lumipulse	≤ 512.9	≤ 0.07	≥ 419.9	≥ 71.20	≤ 9.24

Cut offs describe the condition to consider a value as PATHOLOGICAL, e.g. ≥ 500 means any values equal or larger 500 to be considered pathological. Note that depending on the analyte pathological values are those larger or smaller than a specific cut-off. For A β 40, there is no meaningful cutoff. Cutoffs should be considered as specific for DZNE research cohorts as they are based on the DELCODE subject's data. Transformed cut offs for the Lumipulse will need re-evaluation once sufficient data is generated, and considered educated guess until that point.

- **Contact:**
 - For questions on this documentation and on use of the data, please contact Frederic Brosseron (Frederic.brosseron@dzne.de)